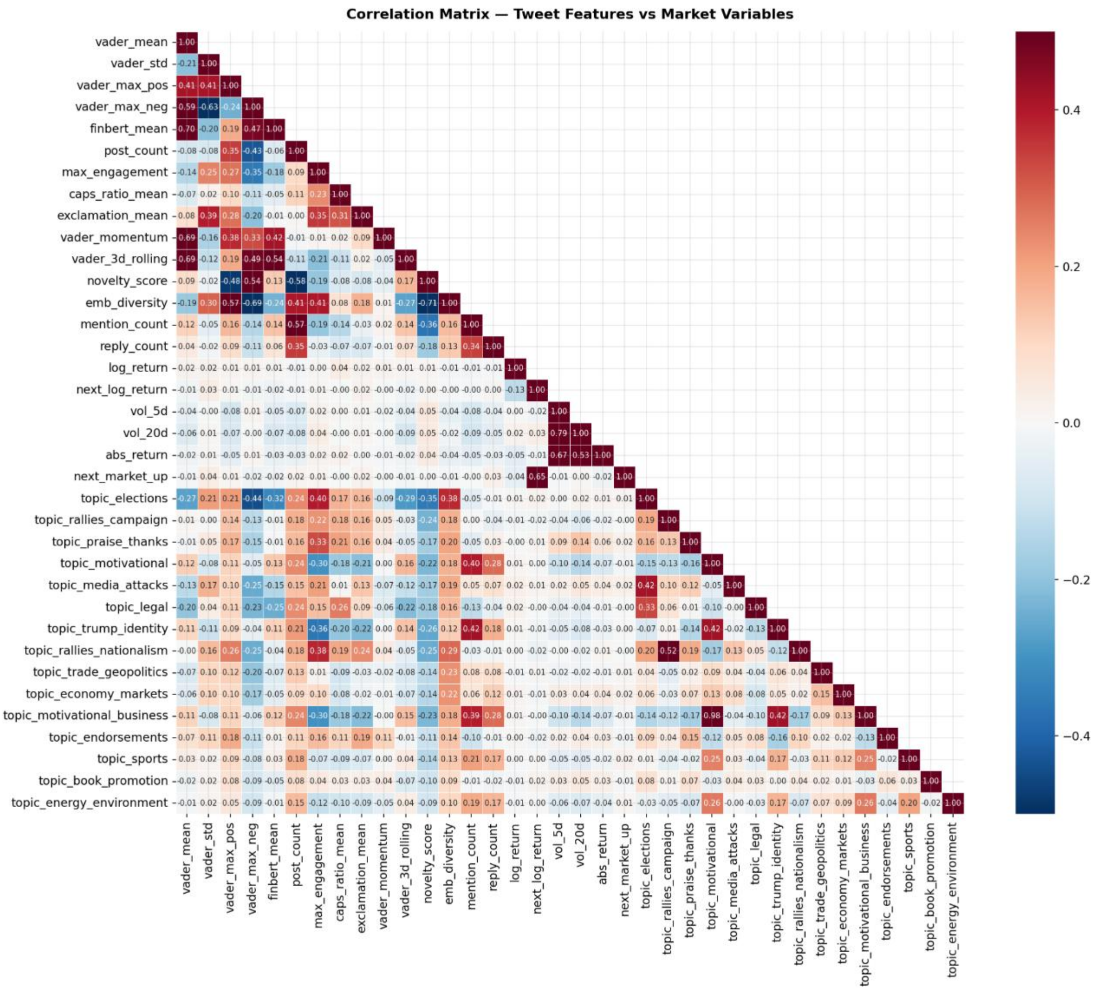
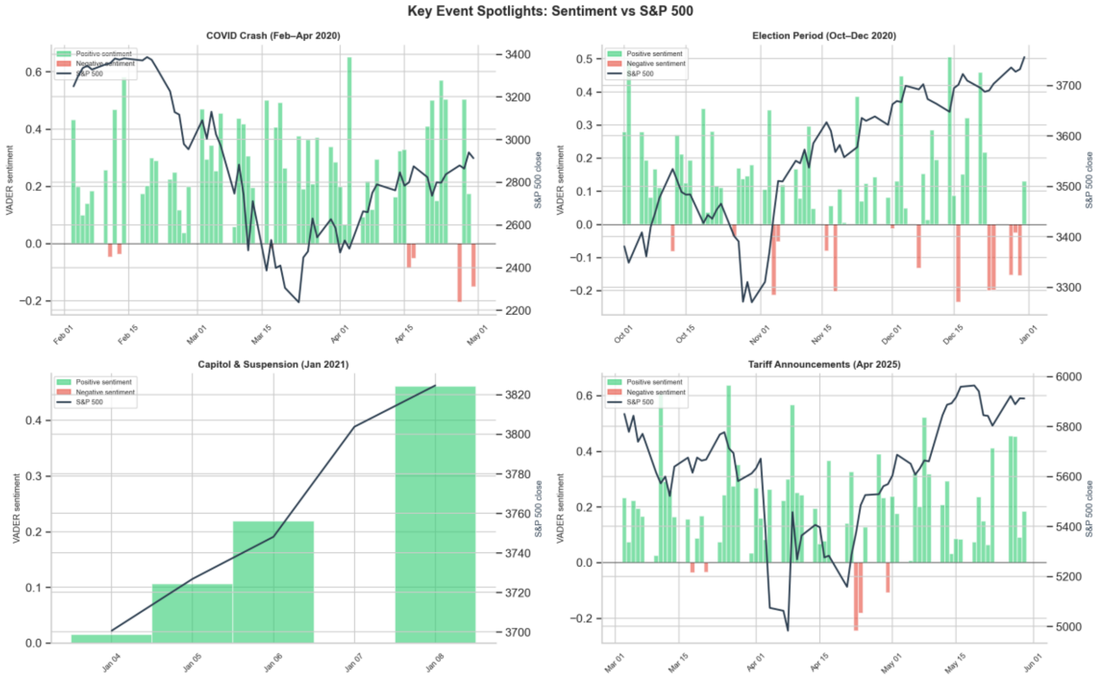
Appendix

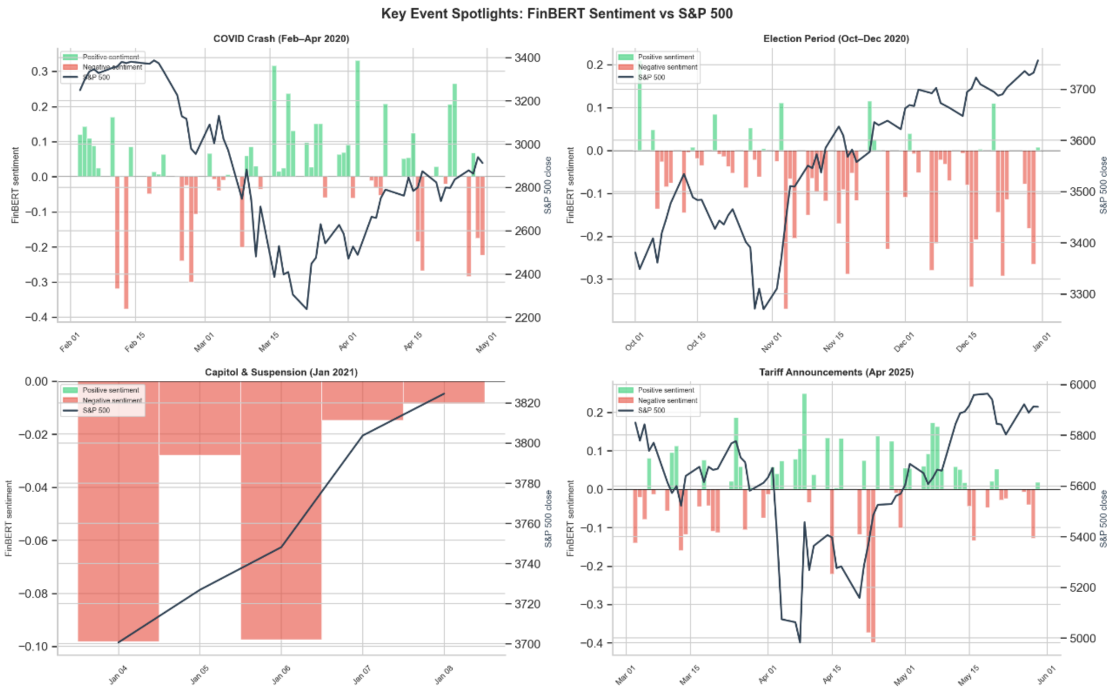
Appendix 1 - Pearson correlation matrix of tweet-level features and S&P 500 market variables.



Appendix 2 – Vader (i) and Finbert (ii).



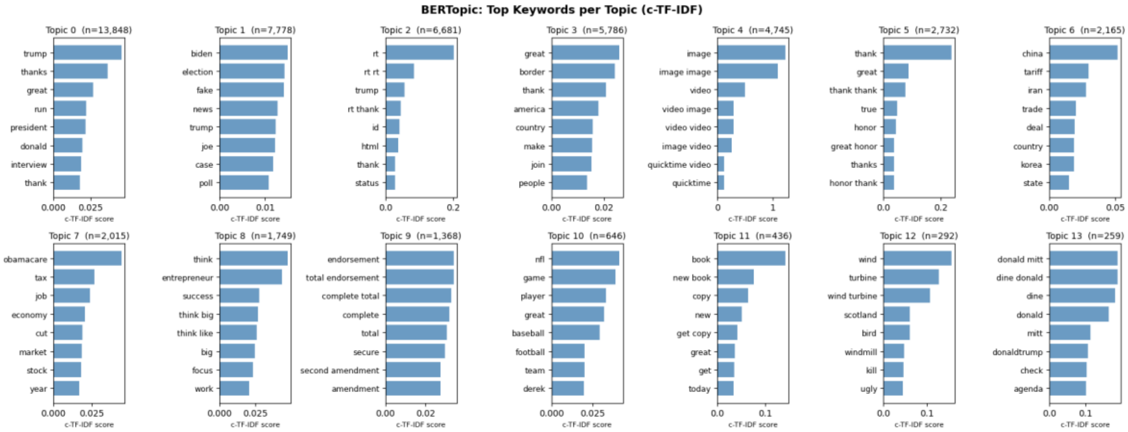
Vader (i)



FinBERT (ii)

Key Event Spotlights: Daily VADER (i) and FinBERT (ii) compound score (bars, left axis) and S&P 500 close price (black line, right axis) across four historically significant periods: the COVID crash (February–April 2020), the 2020 election period (October–December 2020), the Capitol riot and Twitter suspension (January 2021), and the April 2025 tariff announcements.

Appendix 3 - Most relevant words per topic, assessed by BERT-Topic model.

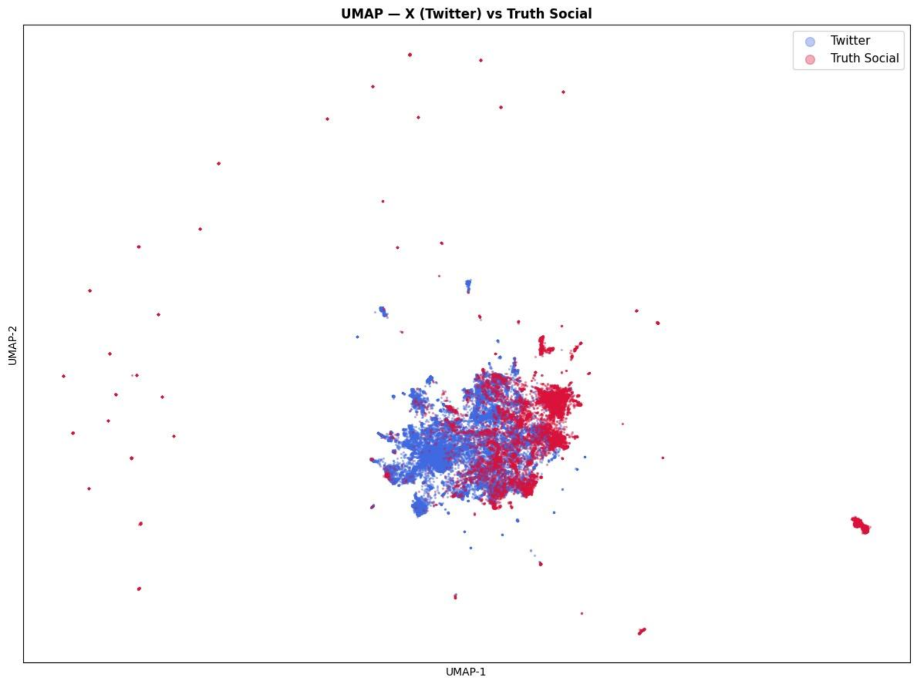


Appendix 4 – Topic keyword flags and representative keyword sets.

Topic Flag	Description	Post Count
topic_trump_identity	Self-referential, personal branding	13,848
topic_elections	Elections, Biden, polls, media narratives	7778
topic_rallies_nationalism	Patriotic rally rhetoric, borders, national pride	5786
topic_praise_thanks	Gratitude, praise, public appreciation	2732
topic_trade_geopolitics	Trade conflicts, tariffs, foreign policy	2165
topic_economy_markets	Economy, taxes, jobs, markets	2015
topic_motivational_business	Business mindset, success, motivational advice	1749
topic_endorsements	Political endorsements, support, Second Amendment	1368
topic_sports	Sports commentary and athlete references	646
topic_book_promotion	Book promotion and sales appeals	436
topic_energy_environment	Criticism of wind energy and environmental themes	292

Appendix 5 – UMAP Projections

i) Next-day S&P500 return



ii) X vs Truth Social

Appendix 6 – Engineered Feature Tables

Preprocessing - Tweet Text

Feature name	Description
date_et	Tweet timestamp converted to US Eastern timezone
post_date	Calendar date of the tweet (date only, no time), in ET
engagement	Sum of favorite_count and repost_count per tweet
tokens	List of cleaned, lowercased, stop-word-free alphabetic tokens extracted from the tweet text
token_count	Number of tokens after cleaning
text_preprocessed	Tokens joined into a single string (cleaned version of the text)
pos_tagged	POS-tagged token list (each token paired with its part-of-speech tag)
lemmas	Lemmatized token list, using POS-aware WordNet lemmatization
lemma_count	Number of lemmas after lemmatization
text_lemmatized	Lemmas joined into a single string (used as input for topic modeling and embeddings)

Preprocessing – Market Features

Feature name	Description
daily_return	Daily percentage return of the S&P 500 close price
abs_return	Absolute value of the daily return (magnitude of market movement)
next_day_return	Next trading day's return (shifted forward by one row; used as prediction target)

Preprocessing - Daily Post Aggregates

Feature name	Description
total_posts	Total number of posts published on a given day
total_engagement	Sum of engagement (likes + reposts) across all posts in a day
avg_engagement	Average engagement per post on a given day
avg_word_count	Average word count across all posts on a given day
twitter_posts	Number of posts published from Twitter on a given day
truth_social_posts	Number of posts published from Truth Social on a given day
quote_posts	Number of quote posts on a given day
repost_posts	Number of reposts on a given day

deleted_posts	Number of deleted posts on a given day
reply_posts	Number of reply posts on a given day

Feature Engineering — Tweet-Level Sentiment (VADER)

Feature name	Description
vader_neg	VADER negativity score for the tweet text (0–1)
vader_neu	VADER neutrality score for the tweet text (0–1)
vader_pos	VADER positivity score for the tweet text (0–1)
vader_compound	VADER aggregate sentiment score, ranging from −1 (most negative) to +1 (most positive)
vader_label	Categorical sentiment label derived from vader_compound: positive, neutral, or negative

Feature Engineering — Tweet-Level Sentiment (FinBERT)

Feature name	Description
finbert_pos	FinBERT probability of positive financial sentiment
finbert_neg	FinBERT probability of negative financial sentiment
finbert_neu	FinBERT probability of neutral financial sentiment
finbert_label	Dominant FinBERT sentiment class (positive, negative, or neutral)
finbert_polarity	FinBERT polarity score computed as finbert_pos − finbert_neg

Feature Engineering — Topic Features (BERTopic)

Feature name	Description
bertopic_id	Numeric topic ID assigned by BERTopic (−1 = outlier/unclustered)
bertopic_label	Human-readable topic label derived from bertopic_id (e.g., topic_elections)
topic_trump_identity	Binary flag: tweet belongs to the "Trump identity" topic
topic_elections	Binary flag: tweet belongs to the "elections" topic
topic_rallies_nationalism	Binary flag: tweet belongs to the "rallies and nationalism" topic

topic_praise_thanks	Binary flag: tweet belongs to the "praise and thanks" topic
topic_trade_geopolitics	Binary flag: tweet belongs to the "trade and geopolitics" topic
topic_economy_markets	Binary flag: tweet belongs to the "economy and markets" topic
topic_motivational_business	Binary flag: tweet belongs to the "motivational and business" topic
topic_endorsements	Binary flag: tweet belongs to the "endorsements" topic
topic_sports	Binary flag: tweet belongs to the "sports" topic
topic_book_promotion	Binary flag: tweet belongs to the "book promotion" topic
topic_energy_environment	Binary flag: tweet belongs to the "energy and environment" topic
topic_mitt_dinner	Binary flag: tweet belongs to the "Dinner with Mitt Romney" topic
rt_noise	Binary flag: retweets (reposts)
image_noise	Binary flag: tweets that act as image legends

Feature Engineering — Structural / Stylistic Features

Feature name	Description
hour	Hour of the day the tweet was posted (US Eastern time, 0–23)
weekday	Day of the week (0 = Monday, 6 = Sunday)
caps_ratio	Fraction of alphabetic characters written in uppercase
exclamation_count	Number of exclamation marks (!) in the tweet
question_count	Number of question marks (?) in the tweet
url_flag	Binary flag: tweet contains at least one URL
log_engagement	Log-transformed engagement score: log1p(favorites + reposts)
is_premarket	Binary flag: tweet was posted between 4:00 and 9:29 AM ET (pre-market hours)
is_market_hours	Binary flag: tweet was posted during regular market hours (9:30 AM–4:00 PM ET)
is_afterhours	Binary flag: tweet was posted after 4:00 PM ET (after market close)